

Understanding calibration

Connect the app’s predictions and accuracy regions to the SDM research papers.

Reexpress two trains an SDM activation (a lightweight neural network adaptor) over cached feature vectors, then constructs an SDM estimator from the labeled data. The *Quick Start* covers the workflow; this brief companion provides a bridge to the research papers for definitions and derivations, and further reading.

Start with a task and a consistent representation

A document represents one classification decision: For example, a question-response pair labeled **incorrect** or **correct**. Prepare vectors outside the app and import labeled data to be used for training and calibration. Evaluation documents may be unlabeled, but assessing accuracy needs known labels.

Use the `embedding` field, `attributes` fields, or both, in which case the app concatenates the embedding with the attributes vector in that order. Keep the feature model/revision, preprocessing, dimensions, and attribute schema consistent across splits and future inputs. The fingerprint records your representation convention; however, these identifiers alone cannot verify compatibility (e.g., that the input vectors come from the same model).

What the adaptor learns and retains

The adaptor learns a compact representation and class decision-boundary from these fixed input vectors. It does not retrain the model that generated them. The selected training split becomes the *support set*. A new document is projected through the adaptor and matched against this saved support. The predicted class comes from the adaptor’s class scores. See [ACL paper](#), §3.1.

The operative signals that can be examined in the app

Signal	Meaning
Similarity, q	Count of consecutive nearest support matches that are correctly classified <i>and</i> agree with the query’s predicted class.
Distance quantile, d	A value in $[0, 1]$ derived from class-wise reference distance distributions, with a min taken over the class-wise distributions to control for class-wise skew in the distribution of outliers. Larger means relatively closer to support. (The raw squared distances $d_0, d_1, \dots$ in the exemplar list instead increase with separation.)
Magnitude, z'	The adaptor’s class scores. Their relative values combine with q and d to produce the SDM activation. The displayed magnitude margin compares the predicted score with the largest other score.
Rescaled Similarity, q'	Combines q with the output of the SDM activation for the predicted-class, incorporating distance and magnitude into the quantity used to partition calibration data.

Definitions: [ACL paper](#), §3.1.2–3.1.5, Eq. 4–6; §3.2.1, Eq. 8.

Select the model and its support

With shuffling enabled, each training iteration repartitions the combined training/calibration pool. The app selects the checkpoint with the lowest class-balanced average calibration SDM loss across eligible checkpoints and iterations. **Best iteration training** and **Best iteration calibration** show the selected partition and define the support and calibration data used by that model. See [ACL paper](#), §3.1.6.

THE ESTIMATOR

From signals to accuracy regions

An SDM activation and an assigned α region are distinct quantities.

Predicted SDM is the output for the SDM activation for the predicted class. The *SDM estimator* requires sufficient rescaled Similarity *and* an activation output passing its predicted-class threshold. A high activation output alone does not establish membership in a high-reliability (HR) region at a given α .

What calibration constructs

For a target α , such as 0.95, calibration searches successively stricter q' cutoffs. Within each candidate subset, it takes a lower quantile of each class's activation on documents labeled as that class (the empirical 5th percentile for $\alpha = 0.95$). These are the class-wise thresholds. It retains a cutoff when all class thresholds reach at least α . The saved region contains that minimum q' ($q'_{\min}$) and the class thresholds. **Calibration summary** in the app displays these values. The documents that survive the $q'_{\min}$ and the class thresholds for a given α are referred to as the *admitted*, or *non-rejected* set. See [ACL paper, §3.2.1, Algorithm 1 and Eq. 9](#).

This seeks to estimate accuracy *among admitted documents*, assessed in two ways:

Grouping in the app	Accuracy within each group
By true class ($y = c$)	Among admitted, labeled documents whose provided label is c , the fraction predicted correctly.
By prediction ($\hat{y} = c$)	Among admitted, labeled documents predicted as c , the fraction predicted correctly.

The aim is for both sets of conditional accuracies to reach *at least* α . Overall (*marginal*) accuracy can conceal a weak class. Rejected documents are outside these admitted groups. Note that class-conditional accuracy here is not recall over the entire unfiltered dataset. See [ACL paper, §3.2](#).

Why the app has several α regions

Alpha resolution specifies a descending search grid. At resolution 0.01, candidates start at 0.99 and continue down to 0.51. After finding a region, the algorithm removes its admitted calibration documents before searching lower levels. A scored document receives the *highest available* region whose criteria it meets. Some requested levels may yield no region. Finer resolution changes the search; it does not guarantee better estimates. See [research note, §2, Algorithm 2](#).

No region (rejected) means no fitted region accepted the document. It still has a predicted class, which can be correct (or incorrect). Lower regions rank cases outside the most conservative HR region for triage; their transfer under distribution shift can be less robust. See [research note, §2.1 and §3](#).

Centroid and DKW lower

Centroid is the app's name for the estimate using the unadjusted distance quantile. **DKW lower** reduces that quantile using a Dvoretzky–Kiefer–Wolfowitz (DKW) band with effective sample size estimated from q' and the calibration data, then recomputes the activation and selection. It can assign a lower region or reject the document. The Document report in the app shows both. See [ACL paper, Appendix A.5, Eq. 11–19](#), and [research note, §2.2](#).

FROM METHOD TO PRACTICE

Assess the estimates on your data

Use admission, conditional accuracy, and support evidence together.

Preserve a separate evaluation set

Calibration data is used to select the checkpoint and fit regions. Keep evaluation data outside the training/calibration pool, with representative labeled examples of every relevant class. Training and calibration accuracy are not independent evaluations. If you use evaluation results to choose settings or revise data, reserve a further untouched set for the final assessment.

Read admission and accuracy separately; they are different quantities

In Analysis, **All selected documents** includes rejected documents that match the current filters. The bordered **HR metrics** section restricts that selection to assigned regions meeting the displayed α cutoff. Changing **Centroid** to **DKW lower** changes which assignment is used.

Admission is the proportion of evaluated documents whose assigned region meets or exceeds the selected α threshold. In the app, its denominator is all scored documents in the *current selection*, including unlabeled rows. Accuracy uses only known labels.

Check **HR accuracy**, **Min conditional**, the class/prediction cells, and their counts. A high percentage supported by very few labeled cases deserves more scrutiny; an absent group is undefined, not zero. The **Regions** table's cumulative mode reports admission, while exact mode reports **% of selection** for one assigned region. Full-selection summaries use all matching scores, even when charts are sampled to 2,000 points for display purposes.

Minimum accepted α also filters the displayed documents. Return to **All HR levels**, or use **Clear filters**, to compare HR admission against the full selection while retaining the HR cutoff. Reliability diagrams summarize raw activation output and are provided for reference; they do not replace evaluation of the estimator's admitted groups.

Choose an operating point and revisit the evidence

Choose an α cutoff appropriate to the cost of errors and the capacity for further review. With a fixed model, estimator, and document selection, a higher cutoff admits no more documents, but measured accuracy on a finite evaluation set need not increase monotonically. Inspect rejected cases and disagreements alongside nearest exemplars, then obtain reliable labels or additional data where needed. Exemplars are context for inspection, not proof that a prediction or label is correct, and tend to become less informative as the epistemic (reducible) uncertainty increases.

Recalibrate the fixed adaptor changes the α search resolution using the selected checkpoint's saved calibration diagnostics. It does not learn from newly uploaded calibration data or edited labels. To incorporate such changes, retrain or continue training with the appropriate sources, then score again. Recalibration also requires rescoreing to update saved region assignments.

Scope and further reading

The method seeks informative HR regions, and conservative behavior on covariate-shifted data. Arbitrary shifts, adversarial inputs, changed label meanings, weak representations, and label errors can still undermine estimates. Check admission and admitted-group accuracy on relevant labeled data. Additional reading: *Similarity-Distance-Magnitude Activations* and *Research Note: Nested Similarity-Distance-Magnitude Estimators*.